

Graph Temporal Convolution Network in Real-Time Gesture Recognition

EECS 242 Final Project

Motivation

Most current gesture control system support limited pre-defined static gestures, while dynamic gesture provides more degrees of freedom for user.

Works have been explore in using CNN variants to recognize such dynamic gestures and achieve favorable results. However, these approach often introduce higher due to the “future seek” to get the complete input context.

Latency can cause performance downgrade, especially in real-time responses systems (for example, using gesture to play video game):

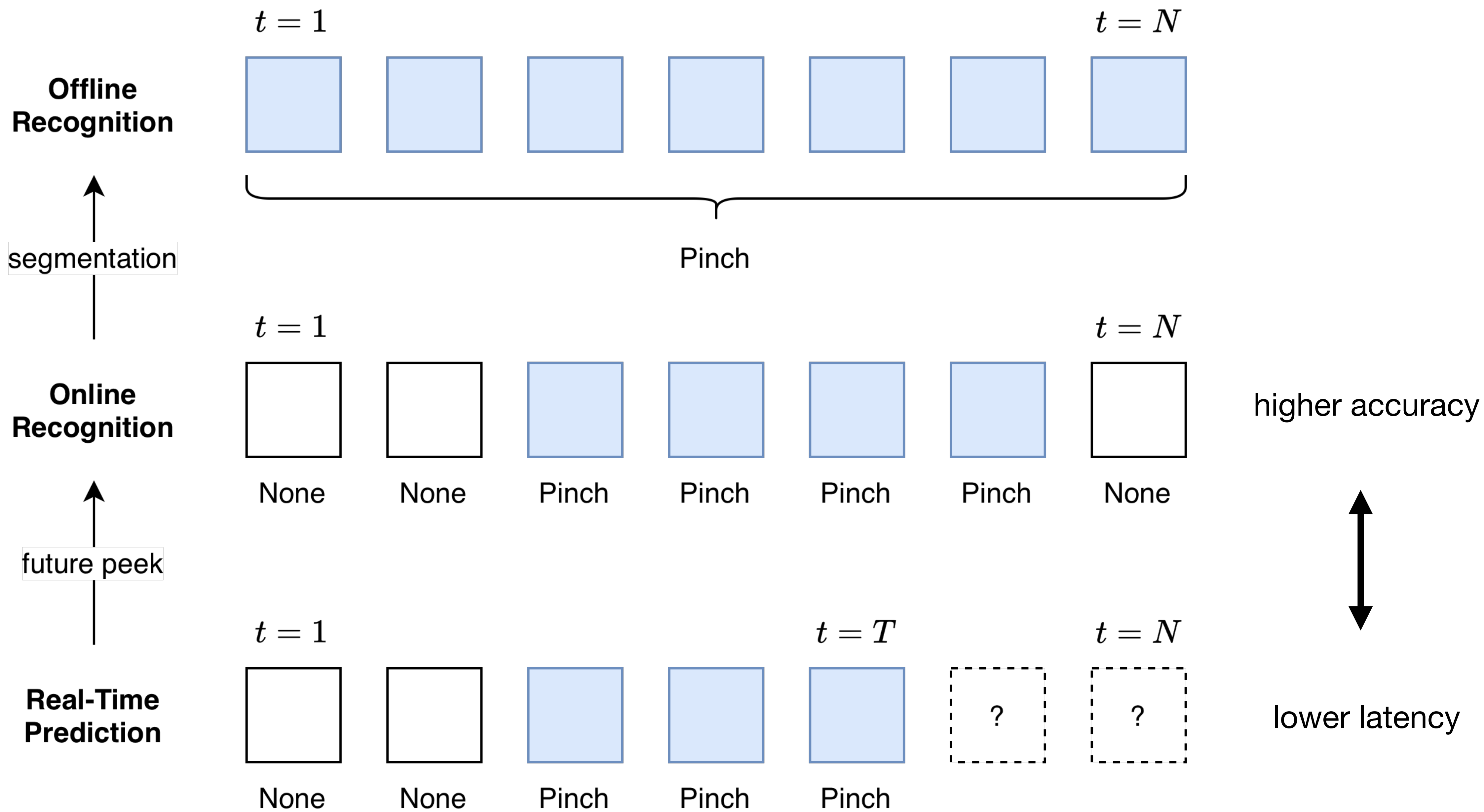
- 30 FPS camera: each frame ~33ms
- latency tolerance in video game: ~150ms



Static Gesture (OK)

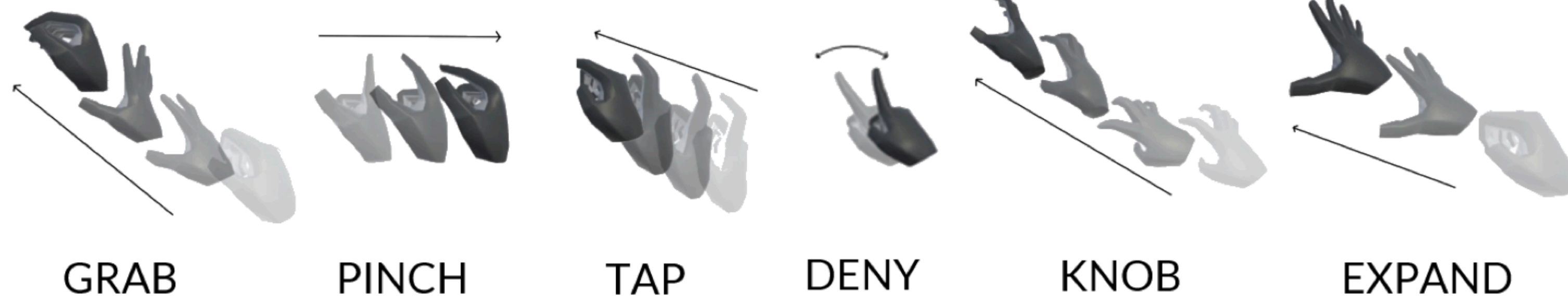
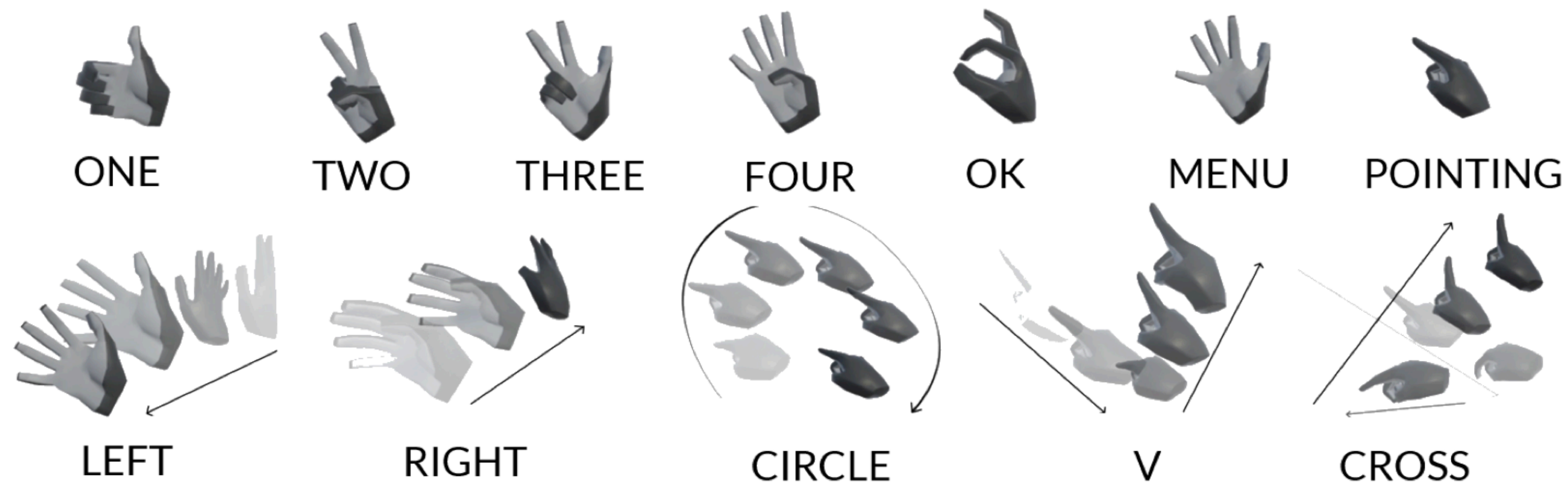


Dynamic Gesture (Pinch)

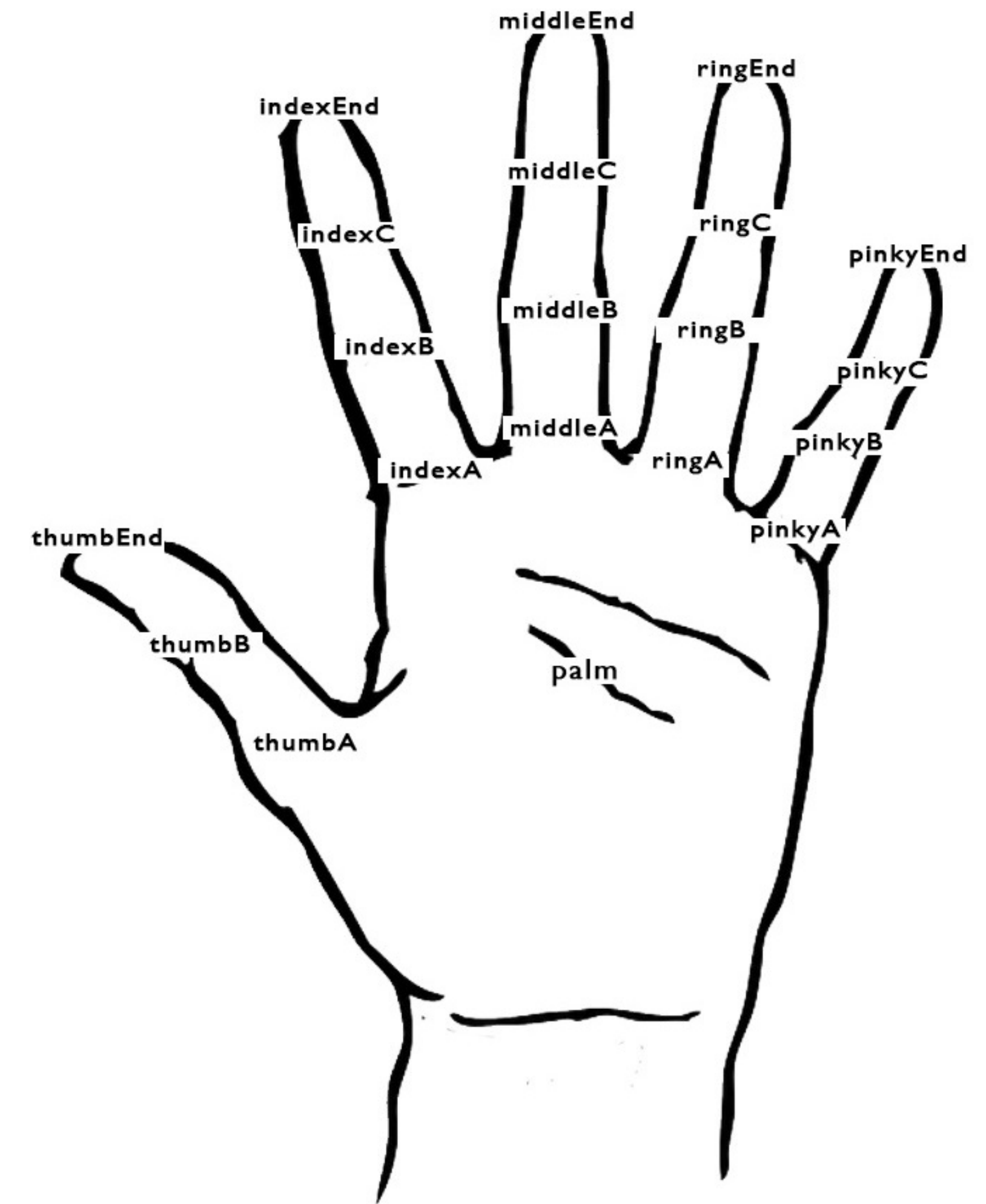


Goal:
Explore the performance of
Graph Temporal-Convolution Network in
Real-Time Gesture Recognition

SHREC 2021 Gesture Dataset



Fine Dynamic Gestures



SHREC 2021 Gesture Dataset

Gesture	Total Occurrences	Total Frames	Avg Duration
NONE	-	168933	-
ONE	24	3461	144.21
TWO	24	3968	165.33
THREE	24	3848	160.33
FOUR	24	3802	158.42
OK	24	3547	147.79
MENU	24	3527	146.96
LEFT	24	666	27.75
RIGHT	24	522	21.75
CIRCLE	24	1370	57.08
V	24	708	29.50
CROSS	24	1106	46.08
GRAB	25	674	26.96
PINCH	23	589	25.61
TAP	24	495	20.62
DENY	24	1978	82.42
KNOB	24	1313	54.71
EXPAND	24	362	15.08

Each sequence has ~**1800 frames**, including 3 to 5 gestures.

Each frame has **20 landmarks**.

Each landmark has 7 data points:
pos(x;y;z); quat(x,y,z,w)

For simplicity, we only consider the position (x, y, z) of Thumb, Index, and Middle in this project.

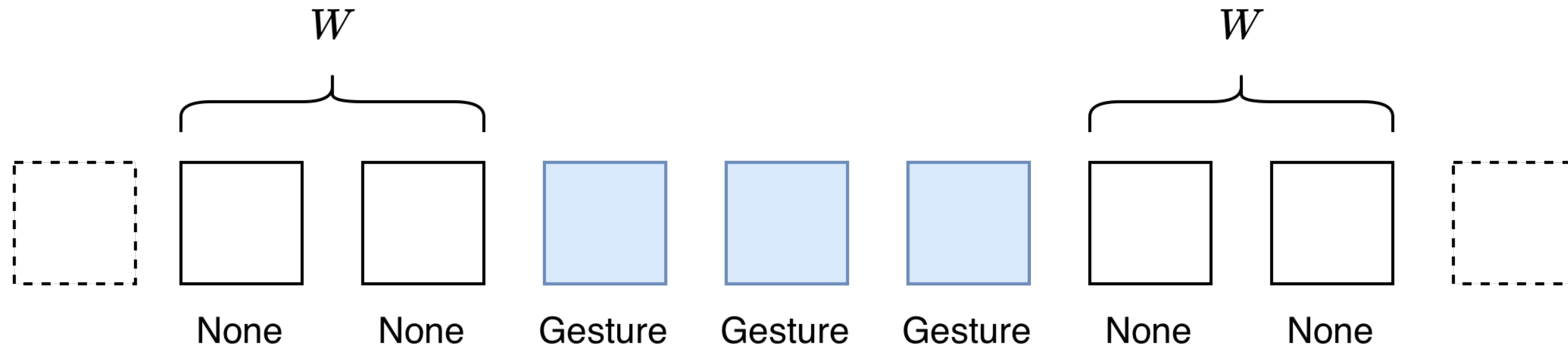
Dealing with NONE Gesture Class

Gesture	Total Occurrences	Total Frames	Avg Duration
NONE	-	168933	-
ONE	24	3461	144.21
TWO	24	3968	165.33
THREE	24	3848	160.33
FOUR	24	3802	158.42
OK	24	3547	147.79
MENU	24	3527	146.96
LEFT	24	666	27.75
RIGHT	24	522	21.75
CIRCLE	24	1370	57.08
V	24	708	29.50
CROSS	24	1106	46.08
GRAB	25	674	26.96
PINCH	23	589	25.61
TAP	24	495	20.62
DENY	24	1978	82.42
KNOB	24	1313	54.71
EXPAND	24	362	15.08

A strong class imbalance where most of the frame being NONE gesture.

We can't directly train the model with all NONE gesture class, since the variance within the NONE gesture is high and might affect model's attention on real gestures.

Data Preprocess



To reduce the number of none gesture class, we extract the pre W frames and post W frames as context, and drop all other (none gesture) frames.

Train set: NONE=2860; GRAB=674; PINCH=589; TAP=468; DENY=1978; KNOB=1313; EXPAND=362;

Test set: NONE=1850; GRAB=453; PINCH=608; TAP=421; DENY=1249; KNOB=1207; EXPAND=358;

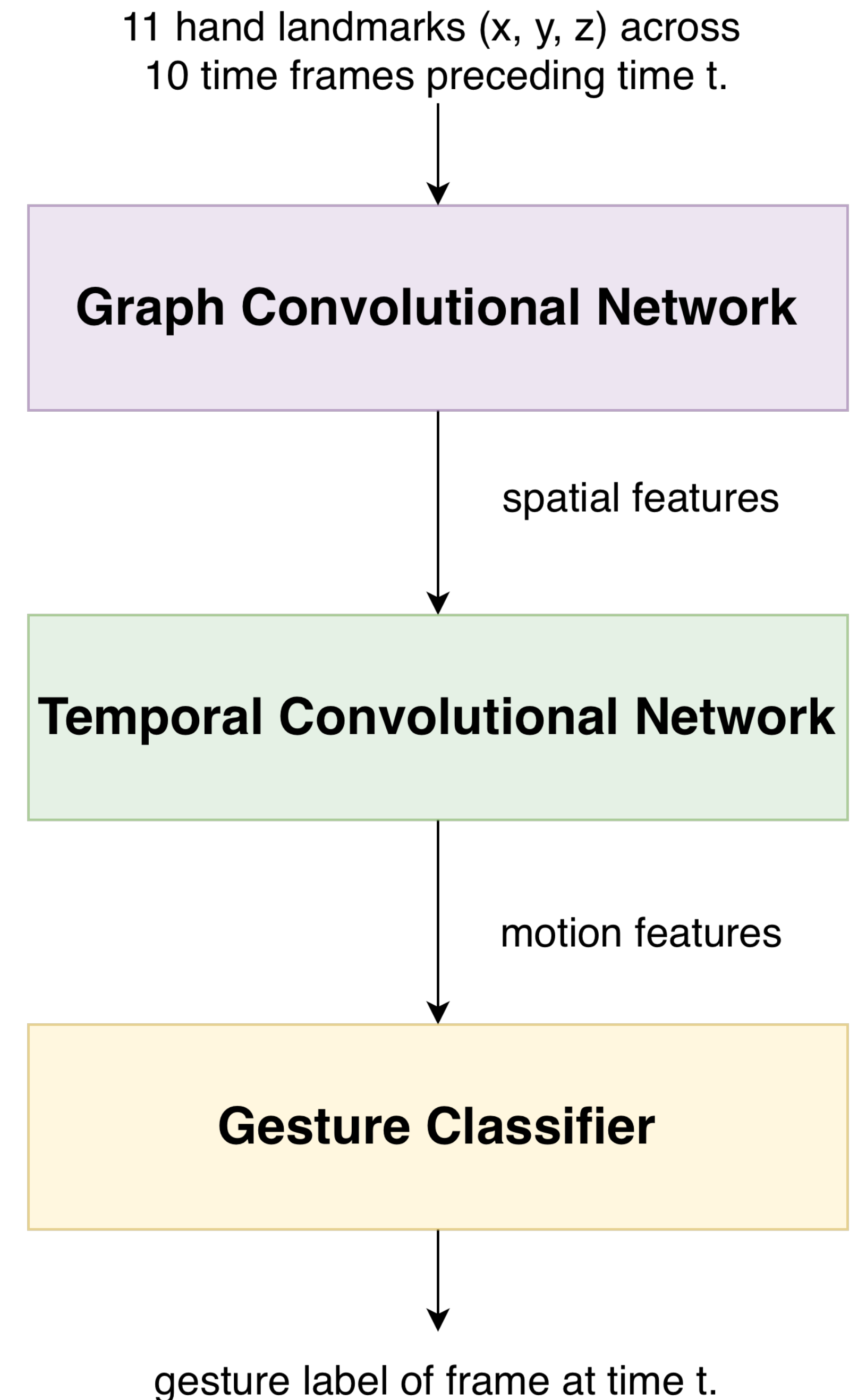
Model

Model Architecture

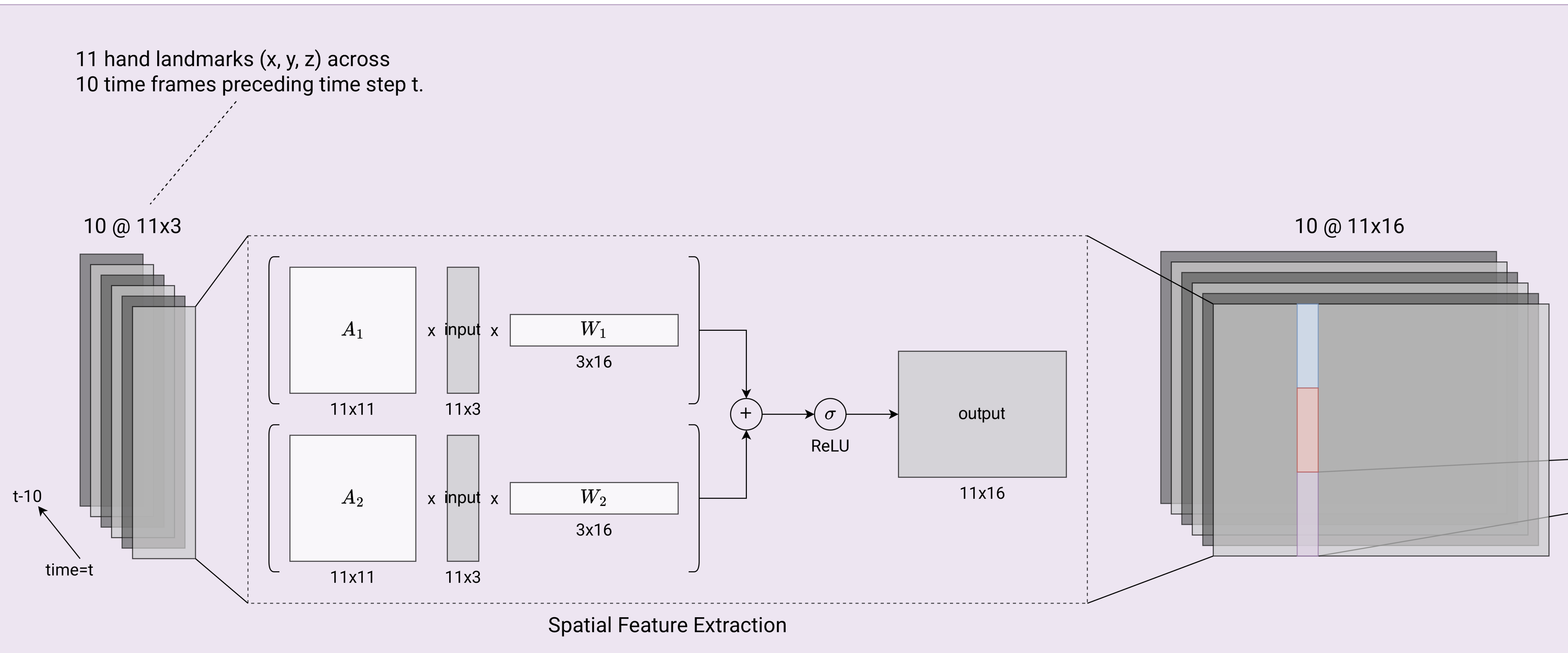
We use a combination of Graph Convolutional Network and Temporal Convolutional Network as our backbone model (GTCN).

To further deal with the none gesture type, we designed three types of classifier to evaluate their performance.

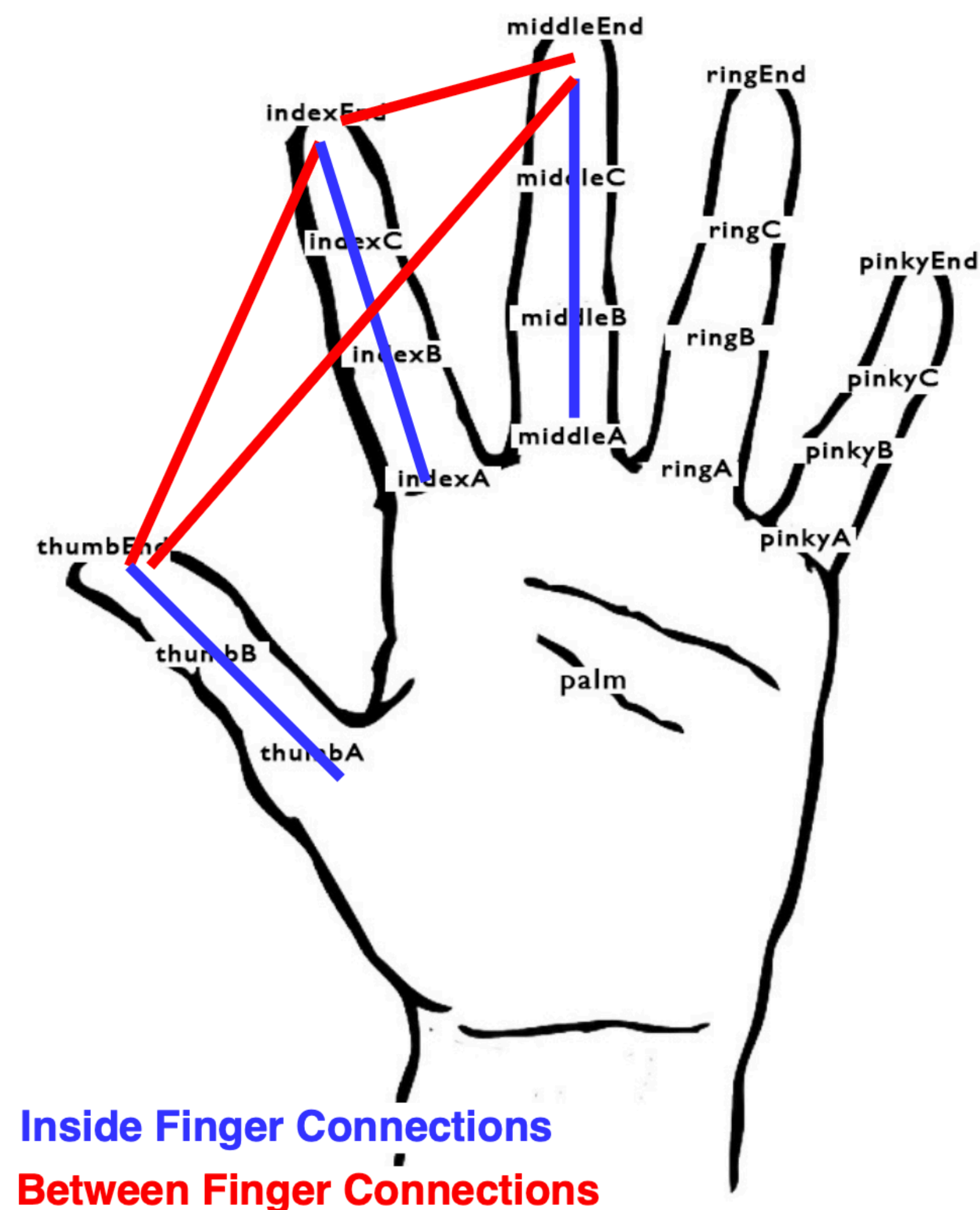
- Model 1: GTCN + Regular Classifier
- Model 2: GTCN + Double Head Classifier
- Model 3: GTCN + Prob Threshold Classifier



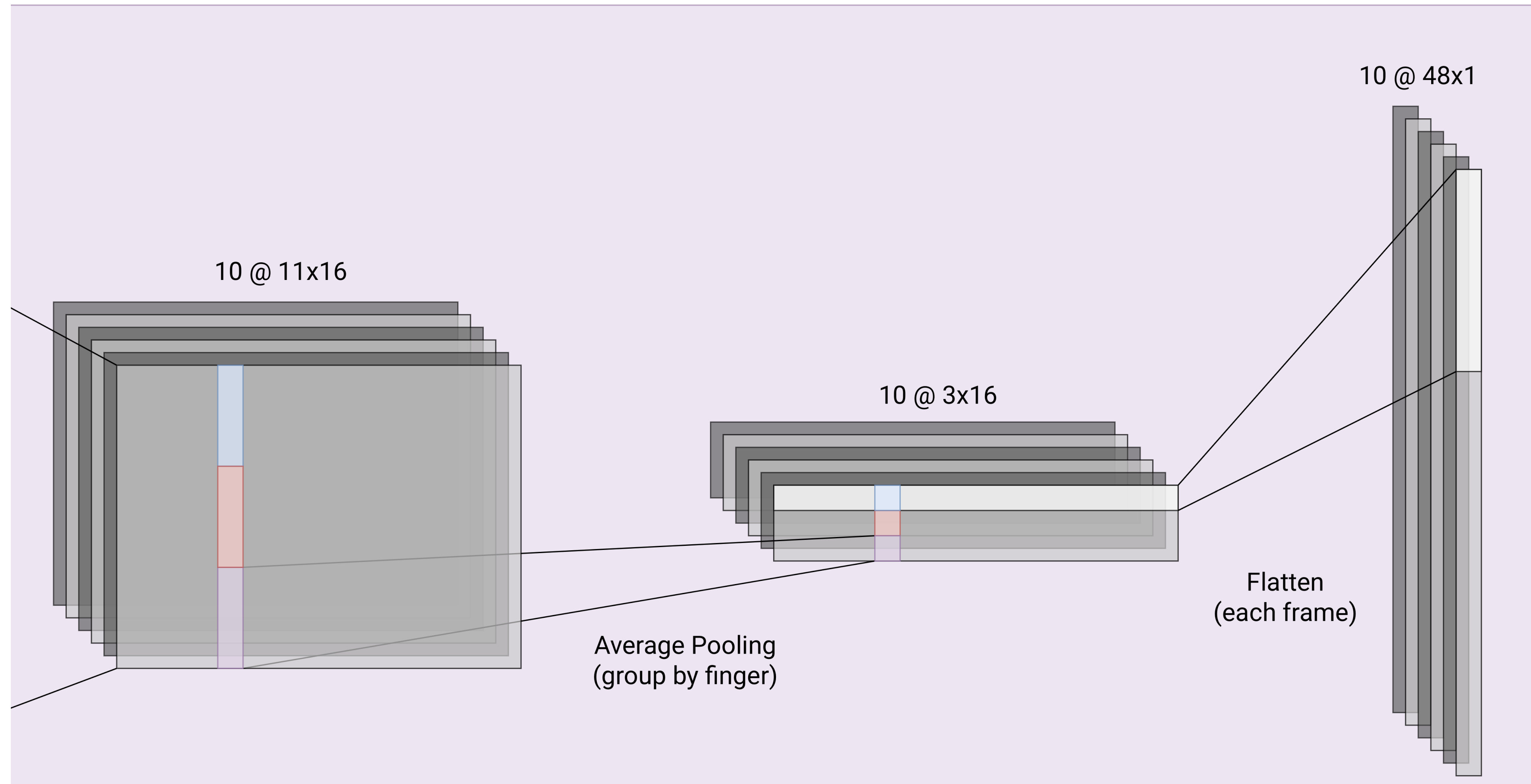
Graph Convolutional Network (1)



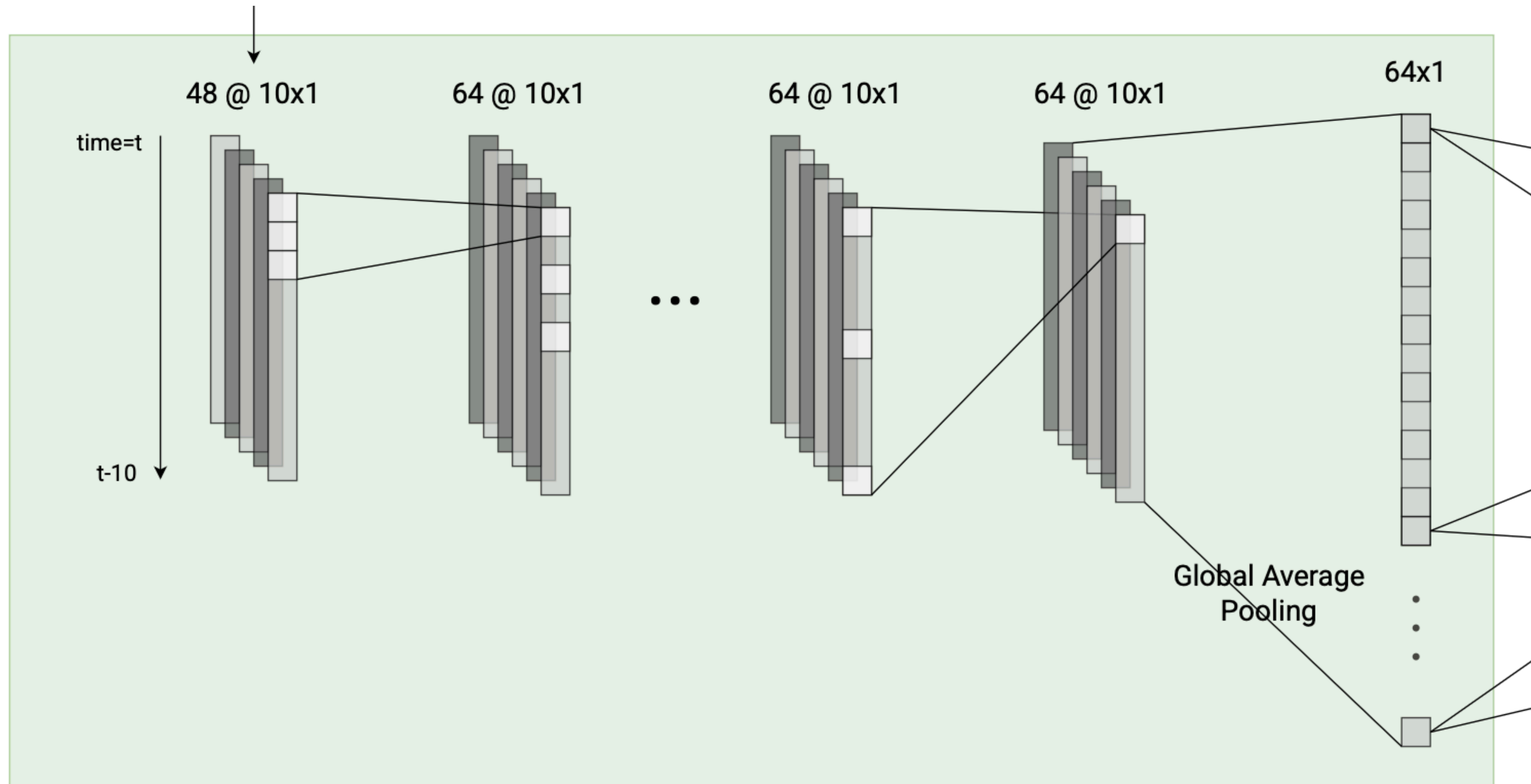
Adjacency Matrix for Finger Connections



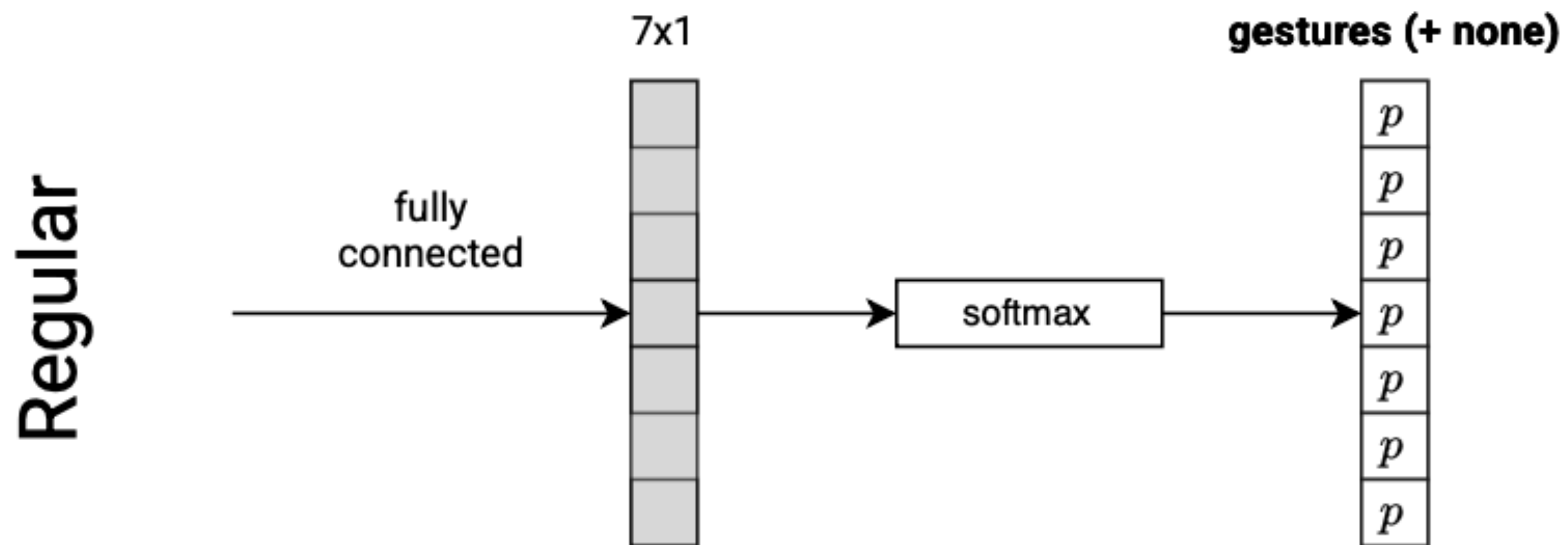
Graph Convolutional Network (2)



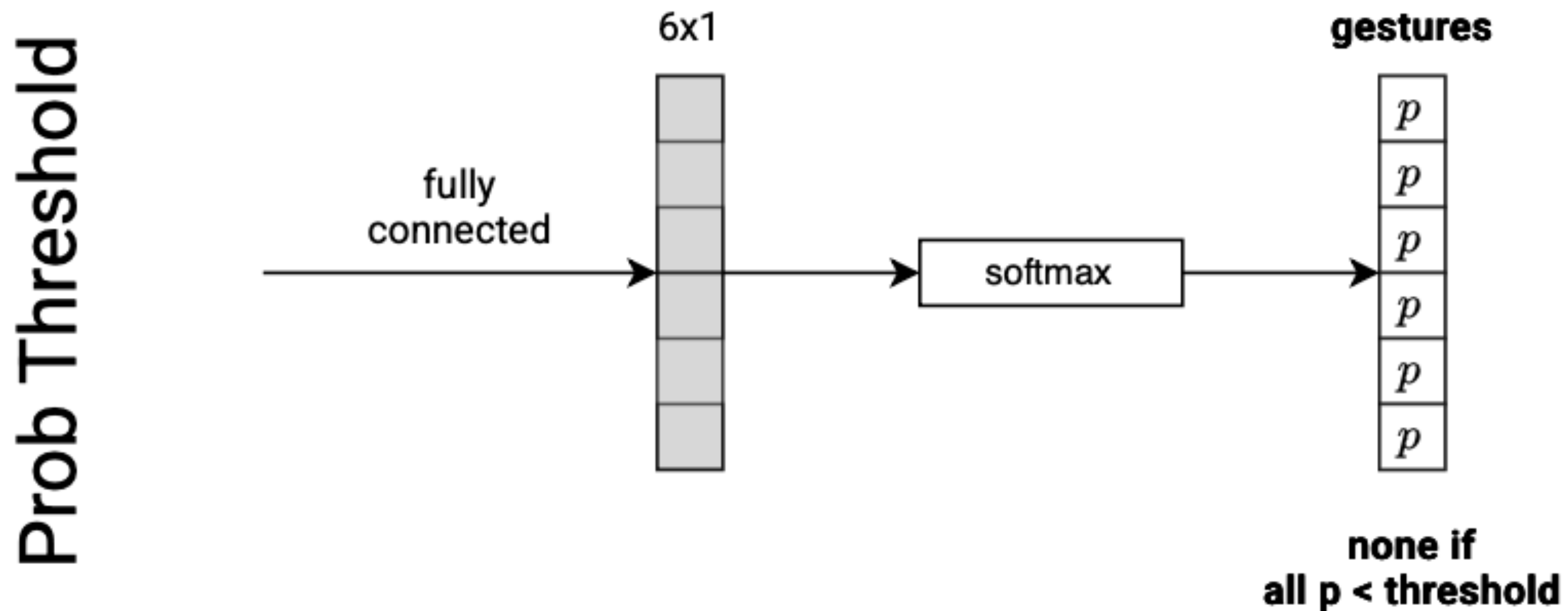
Temporal Convolutional Network



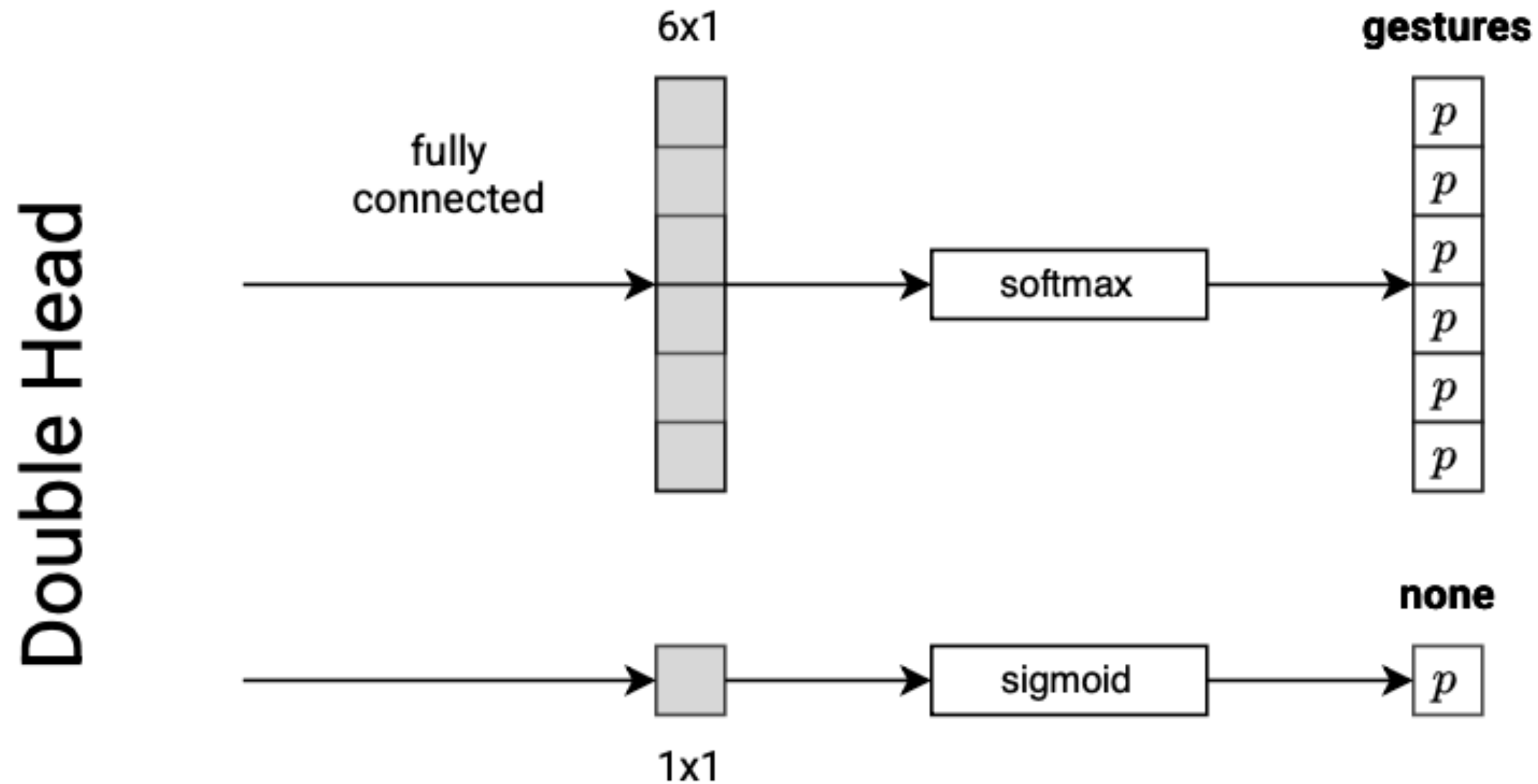
Regular Classifier



Prob Threshold Classifier



Double Head Classifier



$$\text{loss} = \text{loss_gestures} + \text{bce_weight} * \text{loss_none}$$

Optimization & Training

Optimization

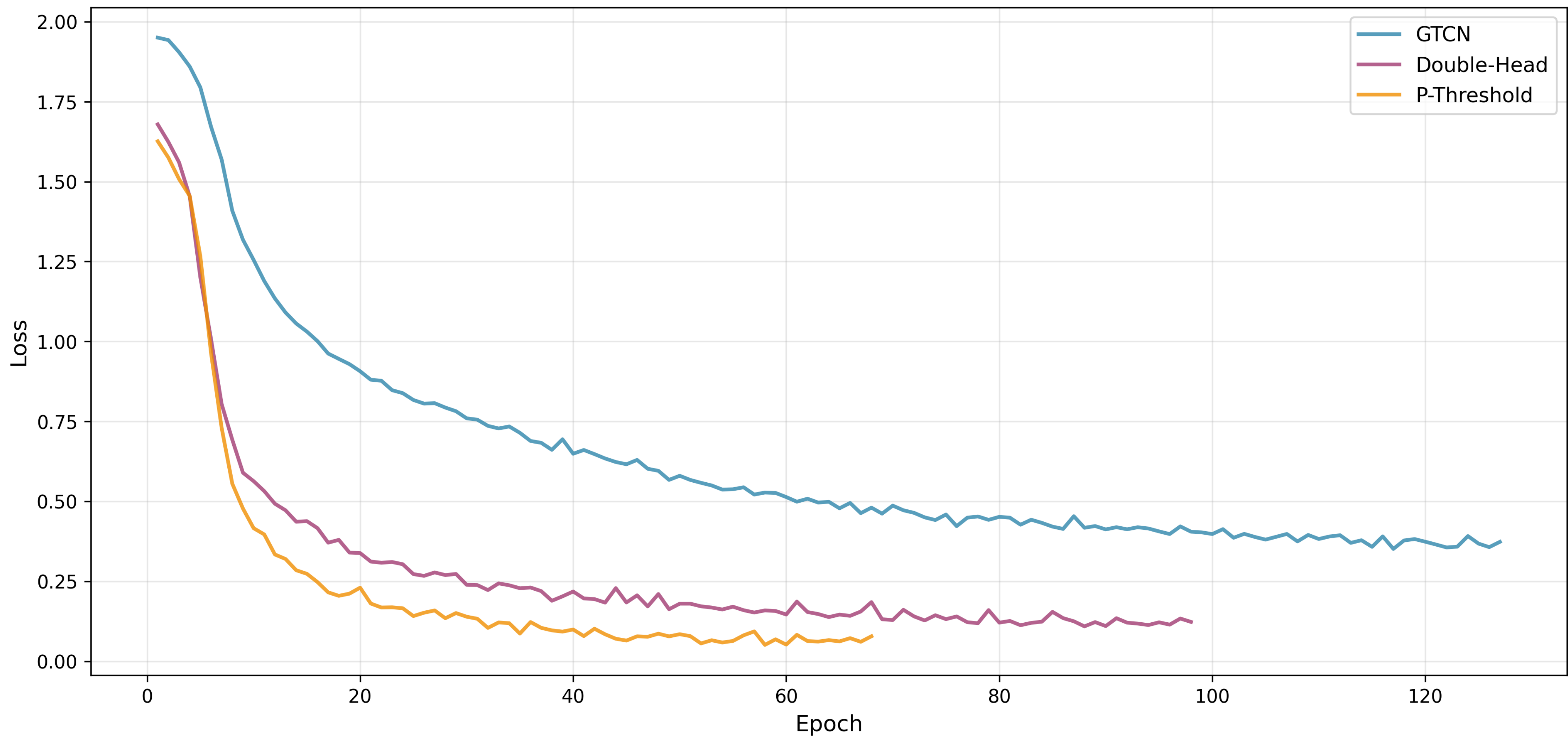
We use Optuna to randomly decide a combination of hyperparameter, and repeat 40 trials to find the best one.

We use 5-fold cross-validation to evaluate the performance of different combinations of hyperparameter.

Optimized Hyperparameter:

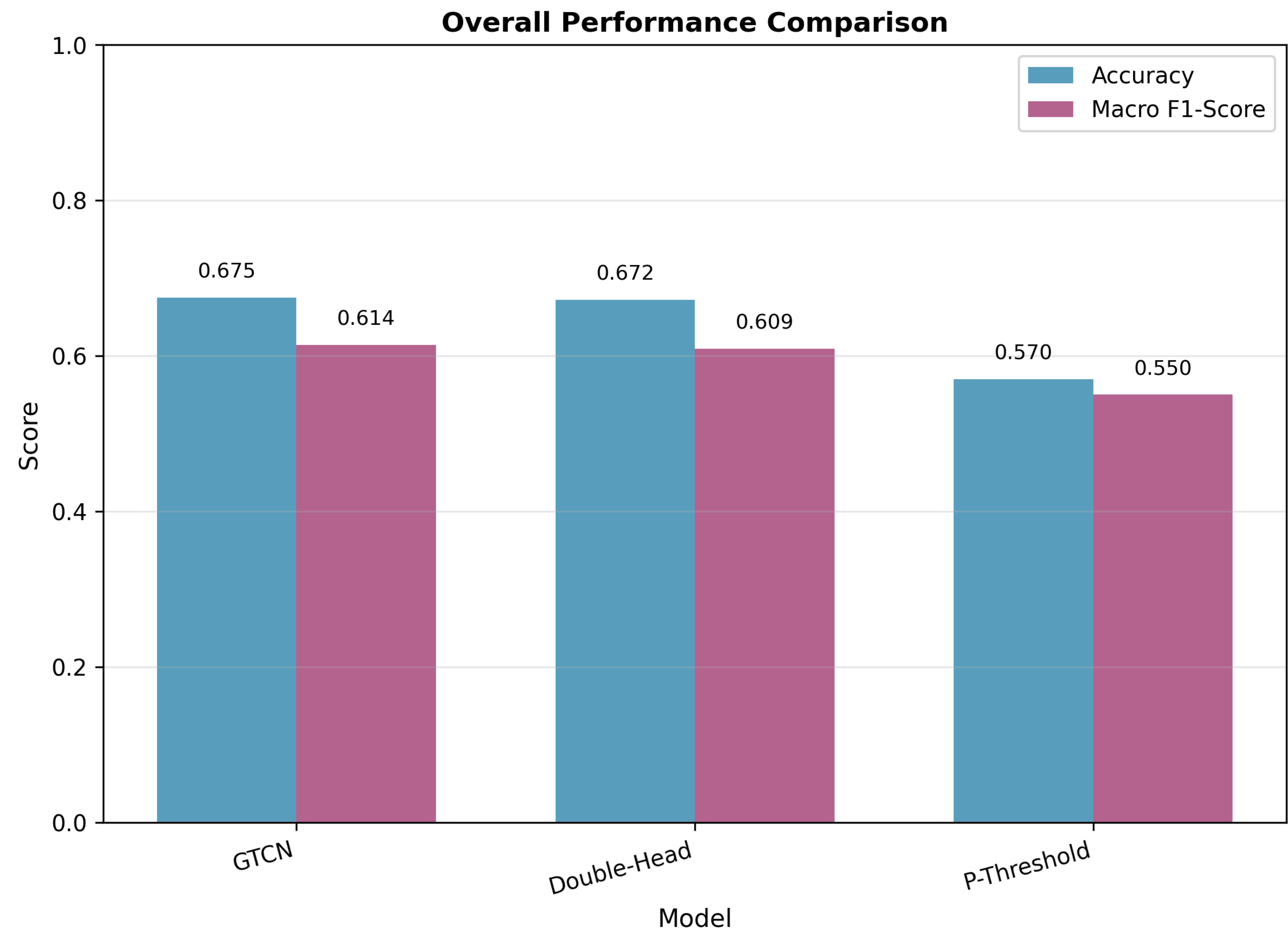
```
GCN_HIDDEN_DIM      = 16
GCN_DROPOUT          = 0.2
TCN_HIDDEN_DIM      = 64
TCN_KERNEL_SIZE     = 5
TCN_DILATIONS        = [1, 2, 4, 8, 16]
TCN_DROPOUT          = 0.2
CLASS_HIDDEN_DIM     = 64
learning_rate        = 0.0033390940624611156
```


Training Curve



Results

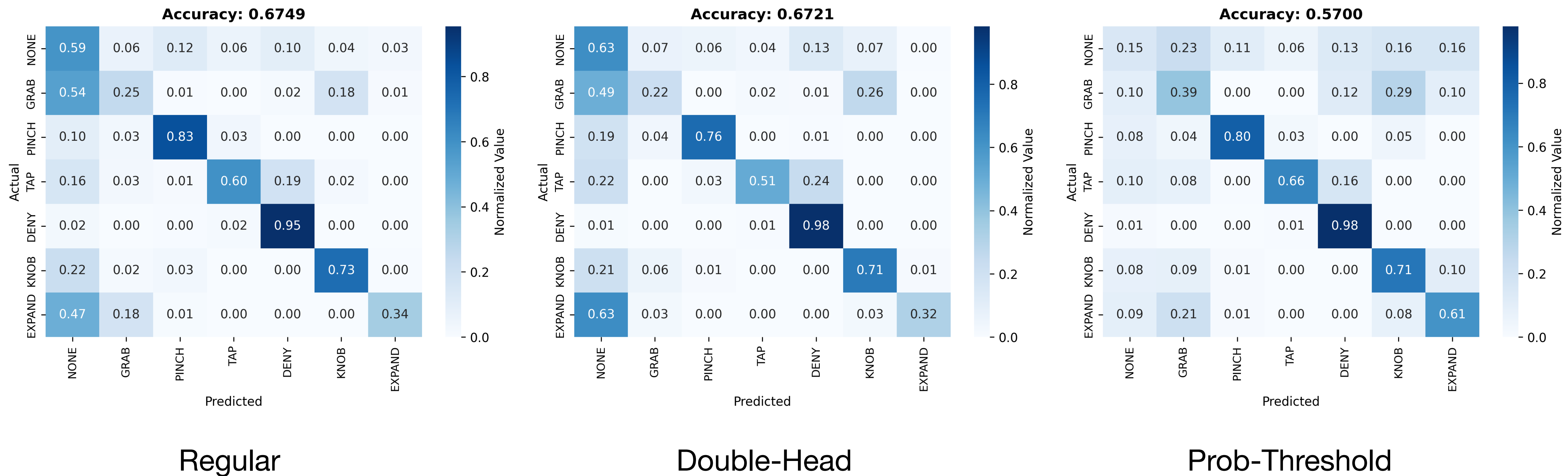
Test Set Performance



Prob-Thresh shows the fastest convergence and lowest training loss, but has lower test accuracy. -> overfitting

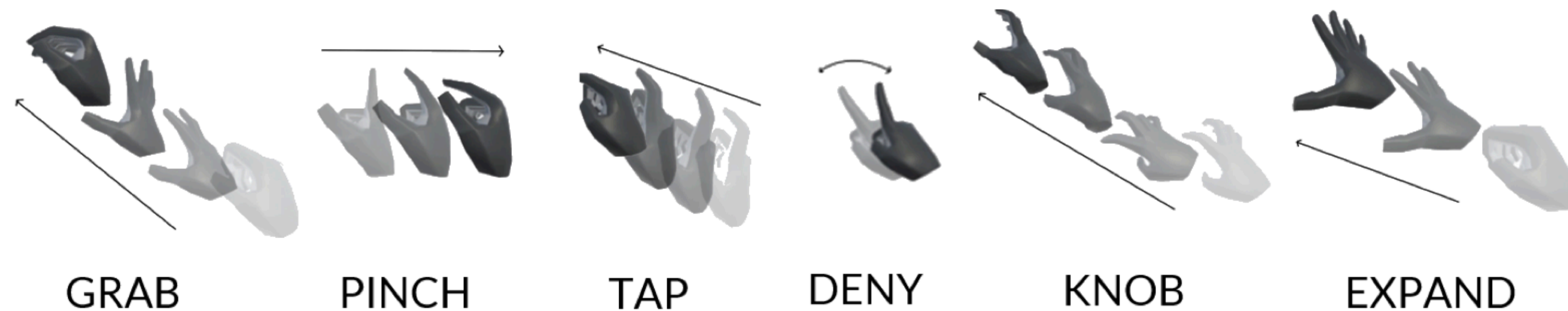
Inference time: 1.6×10^{-4} ms
(per frame)

Confusion Matrix



Observation

- Gestures with higher accuracy—PINCH, TAP, DENY, KNOB—are all performed primarily with the thumb, index, and middle fingers.
 - This indicates that our GTCN architecture is effectively capturing the spatial-temporal patterns of these fingers. (as we only take these three fingers as input.)
- Regular and Double-Head variants exhibit higher misclassification rates where real gestures are mistakenly labeled as NONE, while Prob-Thresh significantly reduces these errors.
 - Prob-Thresh better handles the imbalance in the NONE class by preventing the model from collapsing ambiguous frames into the dominant NONE label.



Future Direction

- **Time-Insensitive TCN**

- Current models rely on a fixed input window length (10 frames). With a dataset recorded at 50 fps, this window captures roughly 200 ms of motion.
- However, in deployment scenarios—such as using a 30 fps camera—the same 10-frame window spans about 330 ms. This mismatch in temporal coverage can significantly affect model performance, since the same gesture may appear stretched or compressed depending on the frame rate.

- **Optimal length of input window**

- How many future frames should we peek to get a balance in latency and accuracy?
- Peeking more frames improves prediction stability and accuracy but increases latency.

QA